WORKED SOLUTION

STEP I 1989, Question 11

Mechanics • Projectile Range Optimisation

STEP I 1989, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q11

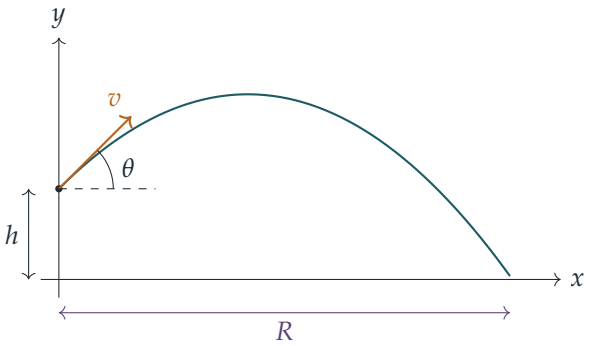
The key idea

The time of flight comes from a quadratic in t: the shot lands when its height returns to zero. Taking the positive root and multiplying by the horizontal speed gives R; the rest is tidying the square root into the required shape. For the second part, "does a small decrease in θ increase R?" is a question about the sign of $\mathrm{d}R/\mathrm{d}\theta$ at $\theta = 45°$, and the symmetry $\sin 45° = \cos 45°$ makes that derivative collapse very neatly.

Solution

The model

We treat the shot as a particle moving under gravity alone, released from the point $(0, h)$ with speed v at angle θ above the horizontal, where $0 < \theta < \frac{\pi}{2}$ and $h > 0$. Take x horizontal in the direction of the throw and y vertically upwards from ground level.



The range

After time t,

$$x = vt\cos\theta, \qquad y = h + vt\sin\theta - \tfrac{1}{2}gt^2.$$

The shot strikes the ground when $y = 0$, that is when

$$\tfrac{1}{2}gt^2 - (v\sin\theta)\,t - h = 0 \quad\Longrightarrow\quad t = \frac{v\sin\theta \pm \sqrt{v^2\sin^2\theta + 2gh}}{g}.$$

Since $h > 0$ the product of the roots, $-2h/g$, is negative, so exactly one root is positive; the minus sign gives the negative one (a time before release), so we take the plus sign. Then

$$\begin{aligned} R = v\cos\theta\, t &= \frac{v\cos\theta}{g}\left(v\sin\theta + \sqrt{v^2\sin^2\theta + 2gh}\right) \\ &= \frac{v^2\sin\theta\cos\theta}{g}\left(1 + \sqrt{1 + \frac{2gh}{v^2\sin^2\theta}}\right) \qquad \text{(taking out } v\sin\theta > 0\text{)} \\ &= \frac{v^2}{2g}\sin 2\theta\left(1 + \sqrt{1 + \frac{2hg}{v^2\sin^2\theta}}\right), \end{aligned}$$

using $2\sin\theta\cos\theta = \sin 2\theta$. This is the required result.

Ask yourself Why is it legitimate to pull $v\sin\theta$ inside the square root as $v^2\sin^2\theta$?

Because $\sqrt{a^2 b} = a\sqrt{b}$ only when $a \geq 0$. Here $0 < \theta < \frac{\pi}{2}$, so $v\sin\theta > 0$ and the step is valid. (For a downward throw it would fail, and the formula as given would be wrong.)

Is 45° the best angle?

For differentiating, the unsimplified form is easier. Write

$$R(\theta) = \frac{v}{g}\cos\theta\,(v\sin\theta + S), \qquad S = \sqrt{v^2\sin^2\theta + 2gh},$$

so that $\dfrac{\mathrm{d}S}{\mathrm{d}\theta} = \dfrac{v^2\sin\theta\cos\theta}{S}$. By the product rule,

$$\frac{\mathrm{d}R}{\mathrm{d}\theta} = \frac{v}{g}\left[-\sin\theta\,(v\sin\theta + S) + \cos\theta\left(v\cos\theta + \frac{v^2\sin\theta\cos\theta}{S}\right)\right].$$

At $\theta = 45^\circ$ put $\sin\theta = \cos\theta = c = \frac{1}{\sqrt{2}}$. The terms $-vc^2$ and $+vc^2$ cancel, leaving

$$\begin{aligned} \left.\frac{\mathrm{d}R}{\mathrm{d}\theta}\right|_{\theta=45^\circ} &= \frac{vc}{g}\left(-S + \frac{v^2c^2}{S}\right) = \frac{vc}{gS}\left(v^2c^2 - S^2\right) \\ &= \frac{vc}{gS}\left(\tfrac{1}{2}v^2 - \tfrac{1}{2}v^2 - 2gh\right) = -\frac{2vhc}{S} = -\frac{\sqrt{2}\,vh}{\sqrt{v^2 + 4gh}}, \end{aligned}$$

where at the last step $S = \sqrt{\frac{1}{2}v^2 + 2gh} = \frac{1}{\sqrt{2}}\sqrt{v^2 + 4gh}$.

Since $v > 0$ and $h > 0$, this derivative is strictly negative. So R is a strictly decreasing function of θ near $45°$, and for a sufficiently small decrease $\delta\theta > 0$ we have $R(45° - \delta\theta) \approx R(45°) + \frac{\sqrt{2}\,vh}{\sqrt{v^2+4gh}}\,\delta\theta > R(45°)$. **Yes: a small decrease in θ will increase R.**

Physically: released from a height, the shot stays in the air a little longer than on level ground, and that extra time is better spent travelling horizontally, so a flatter launch pays off. Only when $h = 0$ does the derivative vanish and $45°$ become optimal.

Checks

- With $h = 0$ the formula gives $R = \frac{v^2}{g}\sin 2\theta$, the familiar level-ground range, and the derivative at $45°$ is 0, as it should be.
- Units: v^2/g is a length, and $2hg/v^2$ is dimensionless.
- Numerically, with $g = 9.8$, $v = 13\,\mathrm{m\,s^{-1}}$ and $h = 2\,\mathrm{m}$: $R(45°) \approx 19.05\,\mathrm{m}$ but $R(40°) \approx 19.10\,\mathrm{m}$. Stepping the motion forward in time reproduces the formula for R, and a finite-difference derivative at $45°$ agrees with $-\sqrt{2}\,vh/\sqrt{v^2+4gh}$ for many random v and h.

Common slips

- Taking the wrong root of the quadratic for t, or forgetting to justify discarding the negative one.
- Reasoning that "$\sin 2\theta$ is maximal at $45°$, so R is too". The bracket also depends on θ (it *decreases* as θ increases), so that argument is incomplete.

Beyond STEP

Setting $\mathrm{d}R/\mathrm{d}\theta = 0$ in general shows that the best launch angle from height h satisfies $\tan\theta = v/\sqrt{v^2+2gh}$, which is a little under $45°$ (about $42°$ for the numbers above); real shot-putters release lower still, because they can push harder at flatter angles. Optimising over a family of trajectories like this leads to the *envelope* of the family, the "parabola of safety", which marks out every point the shot can reach. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops projectile motion and its variations carefully.

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